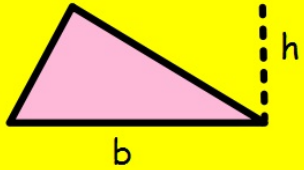
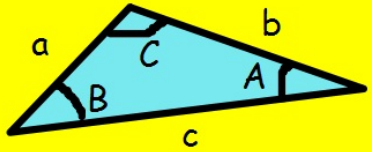
Triangles



$$\text{Area} = \frac{b \times h}{2}$$

$$\text{Area} = \frac{1}{2}ab\sin C$$



Finding Sides and Angles

Is the triangle a right angle triangle?

- yes → are you asked to find a missing side
  - yes → do you know 2 of the sides?
    - yes → Pythagoras Theorem: $a^2 + b^2 = c^2$
    - no → Basic Trigonometry
  - no → Basic Trigonometry
- no → do you know a side and a opposite angle?
  - no → Basic Trigonometry
  - yes → Sine Rule

Side

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Angle

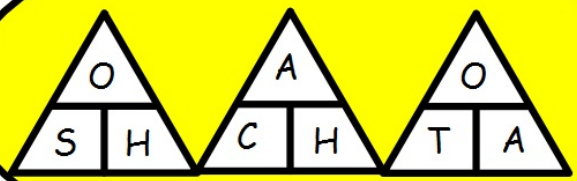
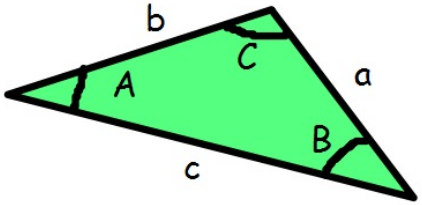
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Sine Rule

For side

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

For angle

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Basic Trigonometry